

OVERVIEW OF VOICE, VIDEO, and DATA COMMUNICATION

Rithesh Kini

Topics to be covered

- I. **Introduction** to **data communication**
- II. **Four fundamental characteristics** a **data communications system**
- III. **Five components** of a **data communications system**
- IV. **Different forms** of **data representation**
(Text, Numbers, Images, Audio, & Video)
- V. **Different modes** of **data flow**

I. Introduction to Data Communication

- **Communication** : means sharing information
 - Locally
 - Remotely (over distance)

I. Introduction to Data Communication

Differences between Communication and Transmission

Communication	Transmission
• It refers to <u>exchange</u> of meaningful information between two communicating devices.	• It refers to the physical <u>movement</u> of information.
• It is a <u>two-way</u> scheme.	• It is a <u>one-way</u> scheme.

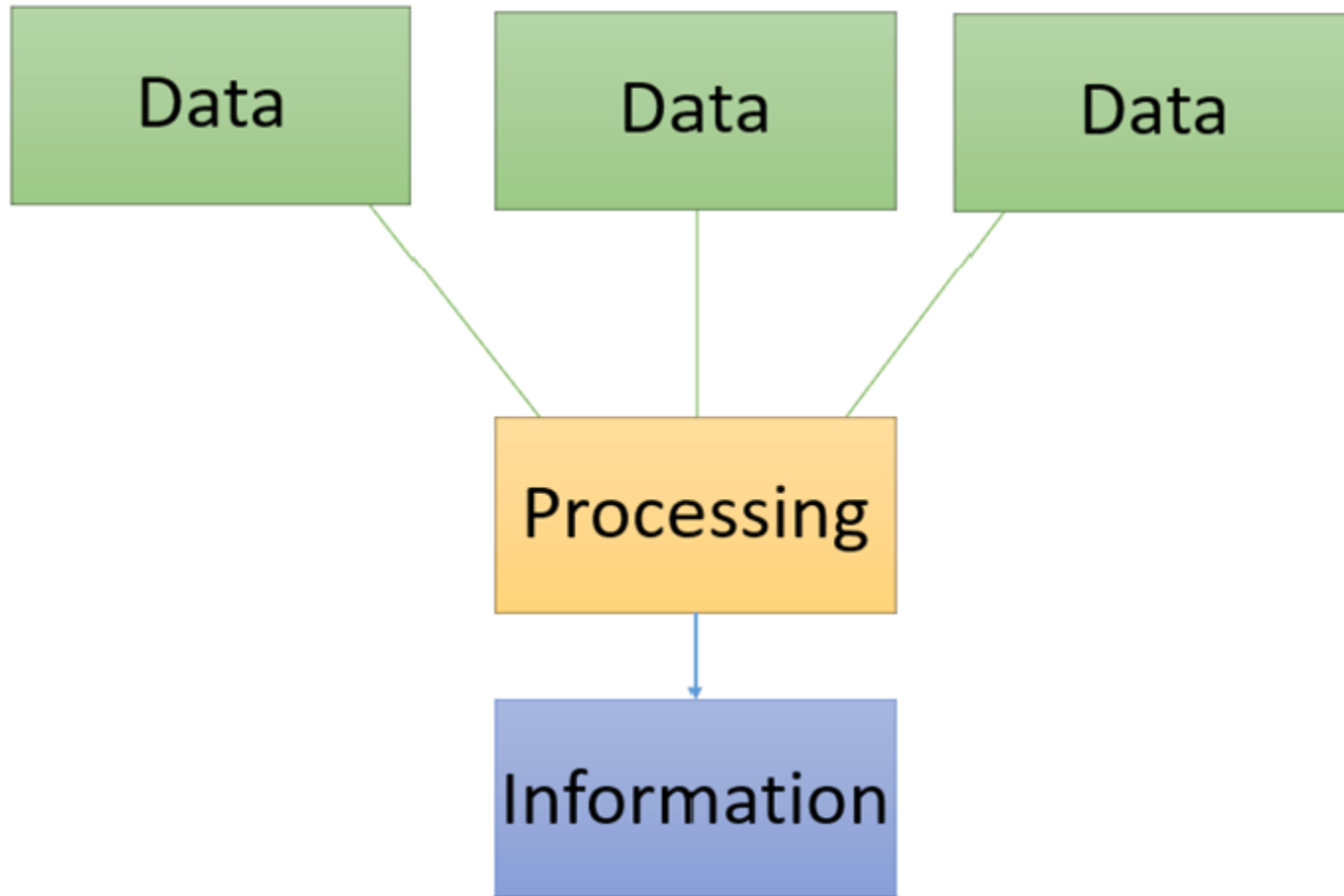
I. Introduction to Data Communication

- **Data** : Refers to information presented in whatever form is agreed upon by the parties creating and using data.

I. Introduction to Data Communication

Data	Information
Data is unorganized raw facts that need processing to make it meaningful	Information is a processed, organized data presented in a given context to make it meaningful and useful
Data comprises facts, observations, perceptions, numbers, characters, symbols, image, etc	Information is processed data that possess context, relevance, and purpose
Data never depends on information	Information is dependent on data
Data is measured in bits and bytes	Information is measured in meaningful units like time, quantity, etc
Data can be structured, tabular data, graph, data tree	Information is language, ideas, and thoughts based on the given data

I. Introduction to Data Communication



I. Introduction to Data Communication

Data

- 100

Information

- 100 miles

Knowledge

- 100 miles is quite a far distance.

Wisdom

- It is very difficult to walk 100 miles by any person, but vehicle transport is okay

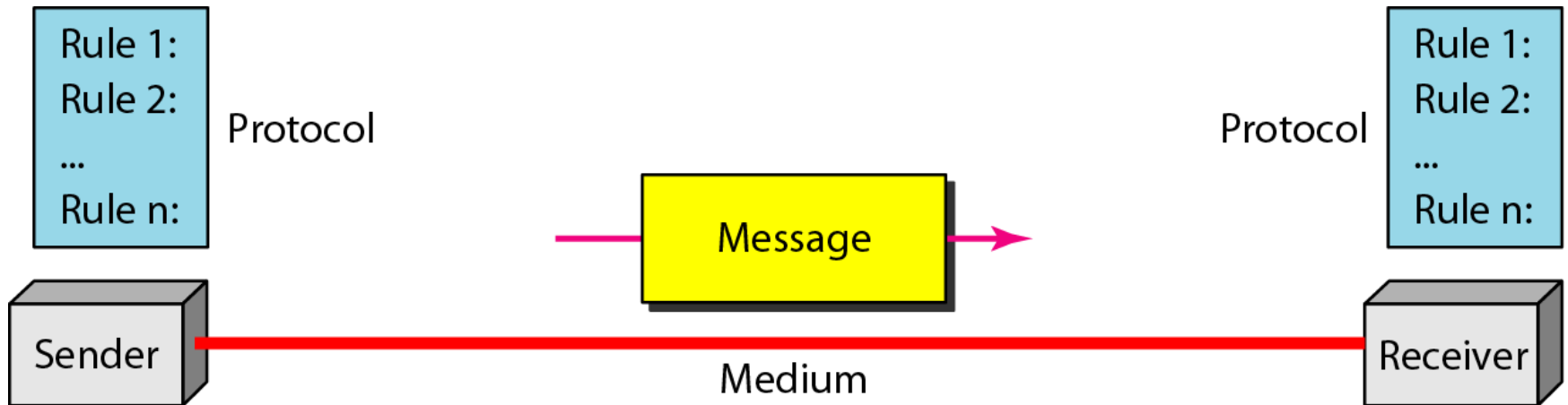
I. Introduction to Data Communication

- **Data communications**: the **exchange** of data between two devices via some form of wired or wireless transmission medium.
- **Communication system** : a combination of **hardware** (physical equipment) and **software** (programs)

II. Four fundamental characteristics of a data communications system

- The effectiveness of a data communications system depends on :
 1. Delivery
 2. Accuracy
 3. Timelines
 4. Jitter (*variation in packet arrival time*)

III. Five components of a data communications system



IV. Different **forms** of **Data Representation**

1. Text

2. Numbers

3. Images

4. Audio

5. Video



HIGH RES IMAGE 300dpi



LOW RES IMAGE 72dpi

DATA FORMATS: Numbers - Binary and Hex

	128	64	32	16	8	4	2	1	Decimal
	b ₇	b ₆	b ₅	b ₄	b ₃	b ₂	b ₁	b ₀	
Val1	0	0	1	1	0	1	0	1	53
Val2	0	1	1	0	1	0	1	0	106
Val3	0	0	0	1	1	0	1	0	26

What is 0010 1011 in decimal?

What is 93 in binary?

Val	Hex		Val	Hex
0000	0		1000	8
0001	1		1001	9
0010	2		1010	A
0011	3		1011	B
0100	4		1100	C
0101	5		1101	D
0110	6		1110	E
0111	7		1111	F

What is 0010 1011 in hex?

DATA FORMATS: Codes - ASCII

Standard ASCII Chart/Table

Dec	Hex	Name	Char	Ctrl-char	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
0	0	Null	NUL	CTRL-@	32	20	Space	64	40	@	96	60	`
1	1	Start of heading	SOH	CTRL-A	33	21	!	65	41	A	97	61	a
2	2	Start of text	STX	CTRL-B	34	22	"	66	42	B	98	62	b
3	3	End of text	ETX	CTRL-C	35	23	#	67	43	C	99	63	c
4	4	End of xmit	EOT	CTRL-D	36	24	\$	68	44	D	100	64	d
5	5	Enquiry	ENQ	CTRL-E	37	25	%	69	45	E	101	65	e
6	6	Acknowledge	ACK	CTRL-F	38	26	&	70	46	F	102	66	f
7	7	Bell	BEL	CTRL-G	39	27	'	71	47	G	103	67	g
8	8	Backspace	BS	CTRL-H	40	28	(	72	48	H	104	68	h
9	9	Horizontal tab	HT	CTRL-I	41	29	)	73	49	I	105	69	i
10	0A	Line feed	LF	CTRL-J	42	2A	*	74	4A	J	106	6A	j
11	0B	Vertical tab	VT	CTRL-K	43	2B	+	75	4B	K	107	6B	k
12	0C	Form feed	FF	CTRL-L	44	2C	,	76	4C	L	108	6C	l
13	0D	Carriage feed	CR	CTRL-M	45	2D	-	77	4D	M	109	6D	m
14	0E	Shift out	SO	CTRL-N	46	2E	.	78	4E	N	110	6E	n
15	0F	Shift in	SI	CTRL-O	47	2F	/	79	4F	O	111	6F	o
16	10	Data line escape	DLE	CTRL-P	48	30	0	80	50	P	112	70	p
17	11	Device control 1	DC1	CTRL-Q	49	31	1	81	51	Q	113	71	q
18	12	Device control 2	DC2	CTRL-R	50	32	2	82	52	R	114	72	r
19	13	Device control 3	DC3	CTRL-S	51	33	3	83	53	S	115	73	s
20	14	Device control 4	DC4	CTRL-T	52	34	4	84	54	T	116	74	t
21	15	Neg acknowledge	NAK	CTRL-U	53	35	5	85	55	U	117	75	u
22	16	Synchronous idle	SYN	CTRL-V	54	36	6	86	56	V	118	76	v
23	17	End of xmit block	ETB	CTRL-W	55	37	7	87	57	W	119	77	w
24	18	Cancel	CAN	CTRL-X	56	38	8	88	58	X	120	78	x
25	19	End of medium	EM	CTRL-Y	57	39	9	89	59	Y	121	79	y
26	1A	Substitute	SUB	CTRL-Z	58	3A	:	90	5A	Z	122	7A	z
27	1B	Escape	ESC	CTRL-[	59	3B	;	91	5B	[	123	7B	{
28	1C	File separator	FS	CTRL-\	60	3C	<	92	5C	\	124	7C	
29	1D	Group separator	GS	CTRL-]	61	3D	=	93	5D	]	125	7D	}
30	1E	Record separator	RS	CTRL-^	62	3E	>	94	5E	^	126	7E	~
31	1F	Unit separator	US	CTRL-`	63	3F	?	95	5F	_	127	7F	DEL

Boolean Logic Operations

AND			OR			X-OR		
A	B	Z	A	B	Z	A	B	Z
0	0	0	0	0	0	0	0	0
0	1	0	0	1	1	0	1	1
1	0	0	1	0	1	1	0	1
1	1	1	1	1	1	1	1	0

Bit shifts

	128	64	32	16	8	4	2	1	Decimal
	b₇	b₆	b₅	b₄	b₃	b₂	b₁	b₀	
Val	0	0	1	1	0	1	0	1	53
Val << 1	0	1	1	0	1	0	1	0	106
Val >> 1	0	0	0	1	1	0	1	0	26

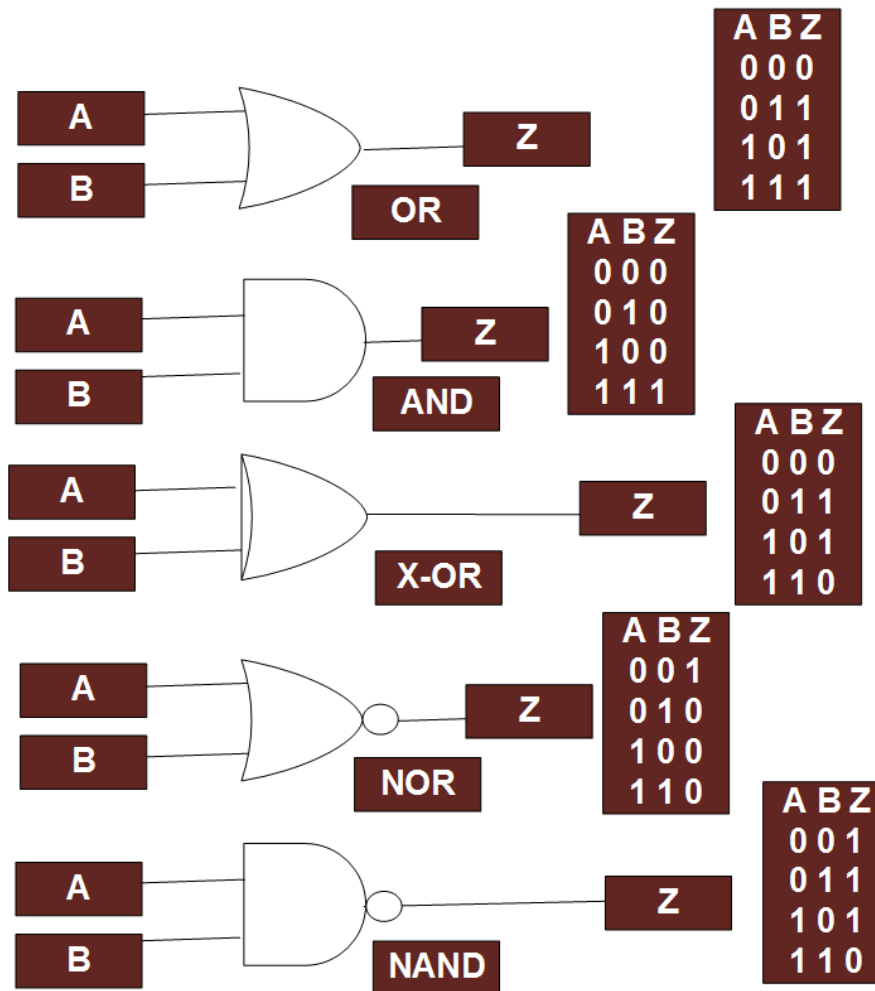
Bit masking

Value	0	0	1	1	0	1	0	1
AND	0	0	0	0	1	1	1	1

	0	0	0	0	0	1	0	1

Result after Masking

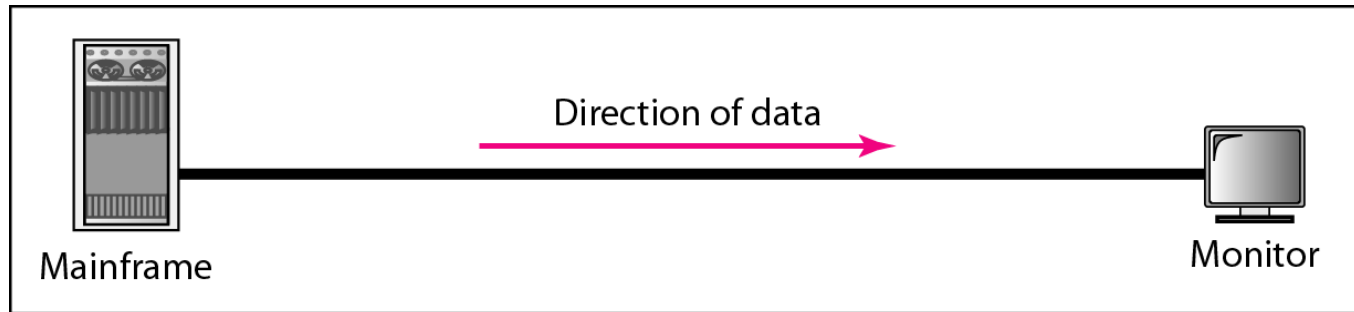
Logic gates



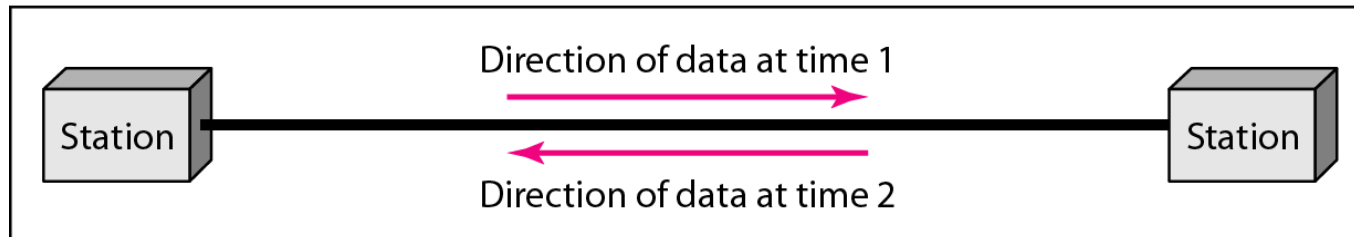
V. Different **Modes of Data Flow**

1. **Simplex** : The communication is unidirectional.
 - E.g., Keyboards(i/p) and traditional monitors(o/p) .
2. **Half-Duplex** : Each station can both transmit and receive, but not at the same time.
 - E.g., walkie-talkies and citizens band radios.
3. **Full-Duplex** (also called **duplex**) : Both stations can transmit and receive simultaneously.
 - E.g., the telephone network.

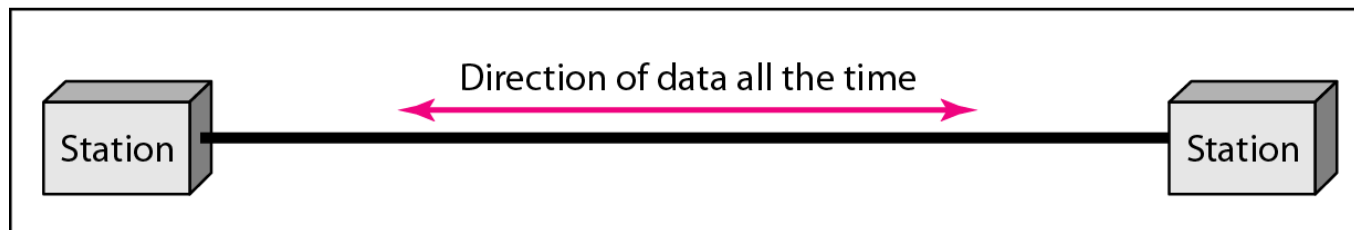
Figure 1.2 *Data flow (simplex, half-duplex, and full-duplex)*



a. Simplex

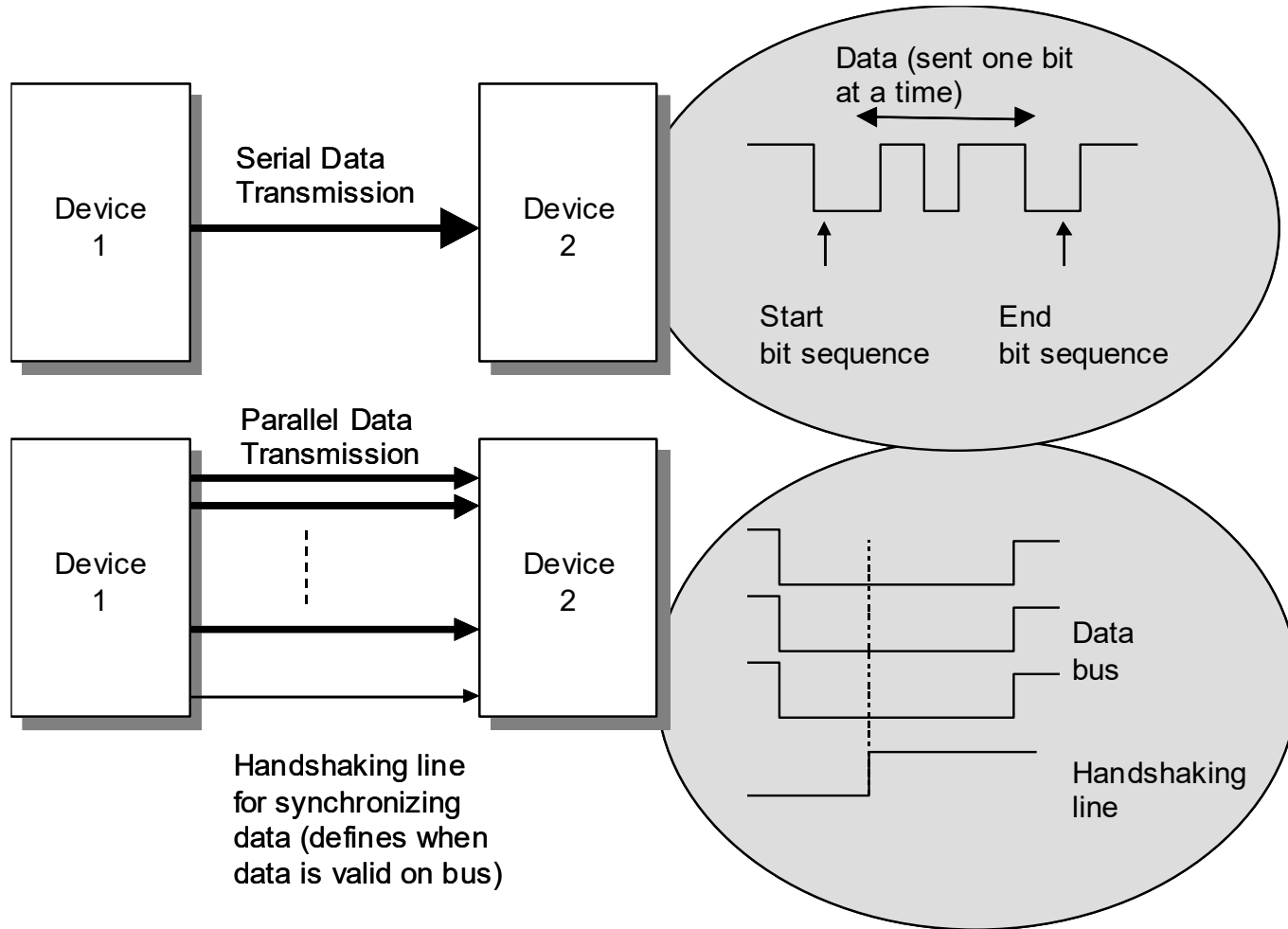


b. Half-duplex



c. Full-duplex

Serial or Parallel?



Thank You